JAVA Programming Project Presentation

KGiSL MicroCollege
Genius as a Service (GaaS)
TECHNICAL TEAM

KGCAS
co-Kreate your Genius

KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

- Project Title: Super market checkout and discount billing system
- Course: java programming Programming Lab
- Student Name & Roll No: Harshini.M 25bcs023
- Department: Computer Science
- Mentor / Guide:Dr.Hemalatha Kanagaraj

TECHNICAL TEAM

JAVA Programming Project Presentation

Introduction

This project is a billing system for supermarkets developed using Java Swing GUI.
It helps to:

- Scan product using barcode
- Automatically fetch product details
- Calculate total bill
- Apply discount
- Generate final bill

Purpose:
To reduce manual billing work and improve accuracy.
Real-life use:
Used in supermarkets, grocery stores, and retail shops.

TECHNICAL TEAM

JAVA Programming Project Presentation

Objective

Main goals of the project:

- To design a GUI-based billing system
- To implement barcode scanning logic
- To calculate total price automatically
- To apply discount percentage
- To generate final bill summary
- To improve programming & problem-solving skills

TECHNICAL TEAM

JAVA Programming Project Presentation

Problem Statement

In many small shops:

- Billing is done manually
- Takes more time
- Chances of calculation errors
- No proper discount system

Solution:

This project provides:

- Fast billing system
- Accurate calculations
- Easy discount application

TECHNICAL TEAM

JAVA Programming Project Presentation

System Requirements

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Storage: 500MB

Software Requirements:

- OS: Windows / Linux
- IDE: NetBeans / Eclipse
- Language: Java
- GUI: Swing

TECHNICAL TEAM

JAVA Programming Project Presentation

System Design

Modules:

1. Barcode Input
2. Product Fetching (HashMap)
3. Billing Calculation
4. Discount Module
5. Bill Generation

Flow:

Barcode → Product → Quantity → Total → Discount → Final Bill

TECHNICAL TEAM

JAVA Programming Project Presentation

Module Description

1. Input Module:
   - User enters/scans barcode
   - Enters quantity
2. Processing Module:
   - Fetch product name & price
   - Calculate total = price × quantity
3. Discount Module:
   - User enters discount %
   - Calculates discount amount
4. Output Module:
   - Displays bill summary
   - Shows final amount

TECHNICAL TEAM

JAVA Programming Project Presentation

Algorithm / Flowchart

- Start
- Initialize total = 0
- Enter barcode
- Check product in HashMap
- Display product name & price
- Enter quantity
- Calculate item total
- Add to overall total
- Apply discount
- Generate final bill
- Stop

TECHNICAL TEAM

JAVA Programming Project Presentation

Implementation

- Techniques used:
- Java Swing (GUI)
- HashMap (store product data)
- Event Handling (button clicks)
- Table Model (display items)
- JOptionPane (input/output)
- Sample products:
- 1001 → Rice → 50
- 1002 → Milk → 30
- 1003 → Oil → 120

TECHNICAL TEAM

JAVA Programming Project Presentation

Output Screens

TECHNICAL TEAM

JAVA Programming Project Presentation

Results & Discussion

System works correctly

Fast billing process

Accurate calculations

Easy to use interface

Challenges faced:

Barcode logic implementation

Handling user input error

Solution:

Used HashMap

Added validation

TECHNICAL TEAM

JAVA Programming Project Presentation

Conclusion & Future Enhancement

- Conclusion:
- Successfully created billing system
- Reduced manual work
- Improved accuracy
- Future Enhancements:
- Add real barcode scanner device
- Connect with database (MySQL)
- Add GST calculation
- Print bill option 🧾
- Add login system

TECHNICAL TEAM

JAVA Programming Project Presentation

References / Acknowledgement

References:

- **Java Programming – E. Balagurusamy**
- **Oracle Java Documentation**
- **Online tutorials**

Acknowledgement:

- **Thanks to my guide**
- **Department staff**
- **Friends for support**

Thank You

- **“From Barcode to Bill in Seconds.”**